

Quantum-Classical Equivalence for AND Functions

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Quantum-Equivalence Conjecture: For a total function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,

$$R(f) \leq \text{poly}(C(f))$$

where

- $R(f)$ = classical randomized communication complexity
- $Q(f)$ = quantum communication complexity (with prior entanglement)

Major open problem in communication complexity. First formally proposed by Shi and Zhu

Our result: For AND functions, $f = g \circ \text{AND}_2$,

$$D(f) \leq \text{poly}(C(f), \log(n))$$

where $D(f)$ = classical deterministic communication complexity.

Thus, Quantum Equivalence Conjecture is true for AND function up to $\text{polylog}(n)$ factors.

This answers an open problem of Razborov [2002].

Why study AND functions?

- Lifted functions form a large class of interesting communication problems. They form a testbed for all major conjectures in communication complexity (e.g, Log Rank Conjecture, Log Approximate Rank Conjecture [now known to be false], Quantum Equivalence Conjecture)

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- Of particular interest is the case when the lifting gadget is as simple as possible, i.e., it has only two bits. In this case, the gadget does not have too much ability to obfuscate the inputs between Alice and Bob.
- There are only two nontrivial gadgets on 2 bits: XOR and AND. Both XOR and AND functions have been extensively studied in the literature.

AND functions

- There is a base function $g : \{0, 1\}^n \rightarrow \{0, 1\}$. Alice has an input $x \in \{0, 1\}^n$, Bob has an input $y \in \{0, 1\}^n$. Their goal is to compute

$$g(x_1 \wedge y_1, \dots, x_n \wedge y_n)$$

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- **Kremer (1996)** proved log-approximate rank lower bounds quantum communication complexity.
- **Lovett, Knop, McGuire, Yuan (2021)** proved the log rank conjecture for AND functions. By an intermediate result in their proof, it became clear that proving the relation conjectured by Burhman and deWolf would also imply the Quantum Equivalence Conjecture for AND functions.

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- They also show the Quantum Equivalence Conjecture for functions of the form $g \circ \text{EQ}_4$.
- We extend their work to show the relation between quantities conjectured by Burhman and deWolf. This implies the Quantum-Equivalence Conjecture for AND functions.

- A result of Kremer ¹:

$$Q^{cc}(F) \geq \text{poly}(\log(\text{rank}^{1/3}(F)))$$

where

$$\text{rank}^{1/3}(F) = \inf\{\text{rank}(G) \mid \|G - F\|_{\infty} \leq 1/3\}$$

¹The result of Kremer only lower bounds clean-quantum communication complexity.[BdW01] observed that it readily implies general quantum communication lower bounds too. Another proof was given by Lee and Shraibman [LS07]

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- It is known that LARC is false in general (**Mande, Suhail, Chattopadhyay, 2019**). Nevertheless, in this work we show LARC is true for AND functions (upto $\text{polylog}(n)$ factors); thereby also establishing the QEC for AND functions.

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In case Yogesh has defined this properly in his previous talk I shall skip the next few slides.

- Any function $g : \{0, 1\}^n \rightarrow \mathbb{R}$ has a unique representation as a multilinear polynomial:

$$g(x) = \sum_{S \subseteq [n]} \alpha_S x_S$$

where $x_S = \prod_{i \in S} x_i$.

Note: This is not the Fourier decomposition where the domain is $\{\pm 1\}$. Also note that we are working over $\mathbb{R}$, not $\mathbb{F}_2$.

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Define sparsity of g , $\text{spar}(g)$, to be the number of nonzero monomials in this representation.

- The conjecture in [BdW01]:

$$\text{rank}^{1/3}[g \circ \text{AND}_2] \approx \text{spar}(g)$$

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- Our main result:

$$\log \text{spar}(g) \leq \text{poly}\left(\log\left(\text{rank}^{1/3}(g \circ \text{AND}_2), \log(n)\right)\right)$$